

Name: _____

Roll No.: _____

Duration: 3 hours

1. (2 points) (MCQ) Denoising Autoencoders and Noise Injection How does a denoising autoencoder utilize input noise injection during its training process?
 - A. By adding noise to the latent representation z to prevent memorization.
 - B. By dropping out a percentage of the hidden layers during reconstruction.
 - C. By feeding a corrupted input $\tilde{x} = x + \epsilon$ to the model and training it to reconstruct the original clean input x .
 - D. By calculating the Mean Squared Error (MSE) only on the noisy features.
2. (2 points) (MCQ) Which method uses second-order derivatives?
 - A. SGD
 - B. Adam
 - C. Newton's Method
 - D. RMSProp
3. (2 points) (MSQ) Which techniques reduce overfitting?
 - A. Dropout
 - B. Data augmentation
 - C. Early stopping
 - D. Increasing training epochs indefinitely
4. (2 points) (MCQ) What is the main advantage of skip connections?
 - A. Reduce parameters
 - B. Improve gradient flow
 - C. Increase computation
 - D. Remove activation functions
5. (2 points) (NAT) Compute output size: input=32, kernel=3, stride=1, padding=1.
6. (2 points) (MCQ) Which optimizer adapts learning rate per parameter?
 - A. SGD
 - B. Adam
 - C. Momentum

- D. Nesterov
7. (2 points) (MSQ) Which are properties of convolution layers?
- A. Weight sharing
 - B. Translation invariance
 - C. Fully connected structure
 - D. Local connectivity
8. (2 points) (MCQ) What does pooling primarily achieve?
- A. Increase resolution
 - B. Reduce spatial size
 - C. Add parameters
 - D. Normalize features
9. (2 points) (NAT) If gradient norm=10 and clipping threshold=5, new norm?
10. (2 points) (MCQ) What is the purpose of BatchNorm?
- A. Increase parameters
 - B. Normalize activations
 - C. Reduce dataset size
 - D. Replace activation
11. (2 points) (MCQ) Which architecture uses skip connections extensively?
- A. VGG
 - B. ResNet
 - C. MLP
 - D. Logistic Regression
12. (2 points) (MCQ) Which loss is used for classification?
- A. MSE
 - B. Cross-entropy
 - C. L1 loss
 - D. Hinge loss only for regression
13. (2 points) (MSQ) Which factors affect training stability?
- A. Learning rate
 - B. Initialization
 - C. Batch size

- D. Dataset size only
14. (2 points) (MCQ) Transfer learning is useful because:
- A. It reduces data requirement
 - B. It increases training time
 - C. It removes need for labels
 - D. It avoids optimization
15. (2 points) (NAT) In a 3-class problem with label smoothing $\epsilon = 0.1$, true class prob?
16. (2 points) (MCQ) Which of the following best describes the primary difference between Standard Momentum and Nesterov Accelerated Gradient (NAG) during steepest descent optimization?
- A. NAG computes the gradient at the current position before applying the velocity update.
 - B. NAG computes the gradient at the "lookahead" position dictated by the momentum term.
 - C. NAG scales the learning rate individually for each parameter based on historical gradients.
 - D. NAG uses a second-order derivative (Hessian) to adjust the step size.
17. (2 points) (MCQ) When training Recurrent Neural Networks (RNNs) on long sequences, exploding gradients can cause numerical instability. The "Gradient Clipping by Norm" technique addresses this by:
- A. Setting the gradient vector strictly to zero if its L2 norm exceeds a predefined threshold.
 - B. Scaling the gradient vector by dividing it by its norm and multiplying by a threshold value, only when the norm exceeds the threshold.
 - C. Clipping the individual elements of the gradient matrix to a fixed minimum and maximum scalar value.
 - D. Adding an L2 penalty to the loss function proportional to the magnitude of the gradients.
18. (2 points) (NAT) A classification task contains 4 classes ($K = 4$). Label smoothing is applied with the parameter $\epsilon = 0.1$. If the true class is class 2, what is the computed smoothed target probability for class 2?
19. (2 points) (MCQ) How does Batch Normalization behave differently during the testing/inference phase compared to the training phase?
- A. It continues to compute the mean and variance of the incoming test mini-batch to normalize the test inputs.

- B. It freezes its learnable scale (γ) and shift (β) parameters, but still computes batch statistics.
 - C. It normalizes the test inputs using an exponentially weighted moving average of the means and variances calculated during training.
 - D. It is entirely bypassed during inference, passing the raw pre-activations directly to the activation function.
20. (3 points) (MSQ) Which of the following statements accurately describe the roles and effects of Data Augmentation and Noise Injection as regularization techniques?
- A. Adding Gaussian noise to the input data during training is mathematically analogous to applying Tikhonov (L2) weight penalty regularization.
 - B. Data augmentation works by synthetically expanding the dataset (e.g., via cropping or flipping), making the model invariant to specific transformations.
 - C. Injecting noise into the hidden activation layers during training guarantees that the loss function becomes perfectly convex.
 - D. Noise injection into the weights themselves encourages the optimizer to find flat minima, which generally generalize better than sharp minima.
21. (2 points) (MCQ) What is the fundamental concept behind Curriculum Training in deep learning?
- A. Alternating the training between the teacher network and the student network on every epoch.
 - B. Freezing hyperparameter updates dynamically based on the validation loss trajectory.
 - C. Presenting training examples in a structured order, starting with the easiest examples and progressively increasing the difficulty.
 - D. Training multiple identical architectures in parallel and averaging their weights to form a stronger ensemble.
22. (1 point) (MCQ) In a multi-label classification problem (e.g., an image containing both "Dog" and "Car"), which combination of output activation and loss function is most suitable?
- A. Sigmoid with Binary Cross-Entropy
 - B. Softmax with Binary Cross-Entropy
 - C. Sigmoid with Categorical Cross-Entropy
 - D. Softmax with Mean Squared Error
23. (1 point) (MCQ) During training, Dropout randomly sets neuron activations to zero. What is its primary purpose?
- A. Increase model capacity
 - B. Reduce overfitting

- C. Speed up training
 - D. Normalize activations
24. (1 point) (NAT) A CNN layer takes input of size 63×63 . It uses kernel size 4, stride 2, and padding 1. After this layer, a second convolution with kernel size 3, stride 2, and padding 0 is applied. Compute the final output spatial dimension.
25. (1 point) (MCQ) For a parameter updated using AdaGrad, what happens if its gradient is consistently zero for many iterations?
- A. Learning rate becomes zero
 - B. Learning rate increases relative to frequently updated parameters
 - C. Parameter is removed
 - D. Gradient becomes non-zero automatically
26. (1 point) (NAT) In RMSProp, the squared gradient moving average is updated as: $v_t = 0.9v_{t-1} + 0.1g_t^2$. If $v_{t-1} = 4$ and $g_t = 6$, compute v_t .
27. (1 point) (MCQ) Layer Normalization differs from Batch Normalization primarily because:
- A. It uses batch statistics across samples
 - B. It normalizes across features within a sample
 - C. It requires large batch sizes
 - D. It cannot be used in transformers
28. (1 point) (MCQ) Why is scaling applied in dot-product attention?
- A. To increase variance of softmax inputs
 - B. To prevent extremely large values leading to small gradients
 - C. To normalize output embeddings
 - D. To reduce computation time
29. (1 point) (NAT) In BatchNorm, given a batch $[2, 4, 6, 8]$, compute the normalized value of 6 (ignore ϵ).
30. (2 points) (MCQ) Match the following architectures with their most defining design choices:

List I (Architecture)	List II (Key Idea)
(A) VGGNet	(1) Introduced skip connections to ease training of deep networks
(B) LeNet-5	(2) Used stacked small filters to approximate larger receptive fields
(C) AlexNet	(3) Early use of convolution + subsampling for digit recognition
(D) ResNet	(4) Popularized ReLU activation and GPU-based training

A. A-2, B-3, C-4, D-1

- B. A-1, B-3, C-2, D-4
 - C. A-2, B-4, C-3, D-1
 - D. A-3, B-2, C-4, D-1
31. (1 point) (NAT) In an attention mechanism, attention weights over 4 tokens are computed as $[2, 1, 0, -1]$ (before softmax). Compute the softmax probability of the first token (rounded to 3 decimal places).
32. Which of the following statements about Batch Normalization are correct?
- A. It reduces internal covariate shift
 - B. It uses running statistics during inference
 - C. It behaves identically during training and inference
 - D. It depends on batch size for stable performance
33. A deep network with ReLU activation is initialized with very small weights. During training many neurons never activate for any input.
- Which phenomenon best explains this behavior?
- A. Vanishing gradients
 - B. Dead neurons
 - C. Exploding gradients
 - D. Overfitting
34. A network uses sigmoid activations and very large positive biases. What is the most likely effect on gradient flow?
- A. Exploding gradients
 - B. Vanishing gradients
 - C. Larger learning rate
 - D. Faster convergence
35. If a transformer model removes positional encoding, what property is lost?
- A. Parallelization
 - B. Order awareness
 - C. Multi-head attention
 - D. Gradient stability
36. If the learning rate is extremely small during training, which phenomenon is most likely?
- A. Divergence
 - B. Extremely slow convergence
 - C. Gradient explosion

- D. Overfitting
37. A model achieves very low training loss but poor validation performance. Which explanation is most plausible?
- A. Underfitting
 - B. Overfitting
 - C. Vanishing gradients
 - D. Gradient clipping
38. During training, gradients suddenly become extremely large for a few iterations before stabilizing.
- Which mechanism can prevent instability?
- A. Dropout
 - B. Gradient clipping
 - C. Data augmentation
 - D. Batch size increase
39. In CNNs, increasing stride primarily affects
- A. Number of parameters
 - B. Spatial resolution of feature maps
 - C. Activation function
 - D. Learning rate
40. Which factors can contribute to vanishing gradients?
- ☐ Sigmoid activation
 - ☐ Deep networks
 - ☐ Small weight initialization
 - ☐ ReLU activation
41. Which mechanisms improve generalization?
- ☐ Dropout
 - ☐ Weight decay
 - ☐ Data augmentation
 - ☐ Increasing training data
42. Which operations increase the receptive field in CNNs?
- ☐ Increasing kernel size
 - ☐ Increasing depth

- ☐ Increasing stride
 - ☐ Batch normalization
43. Which components exist in transformer encoder layers?
- ☐ Multi-head attention
 - ☐ Feed-forward network
 - ☐ Residual connections
 - ☐ Positional encoding
44. Which situations may cause exploding gradients?
- ☐ Large weight initialization
 - ☐ Deep networks
 - ☐ Large learning rate
 - ☐ Batch normalization
45. Which techniques help stabilize training of deep networks?
- ☐ Batch normalization
 - ☐ Residual connections
 - ☐ Gradient clipping
 - ☐ Careful initialization
46. Advantages of CNN architectures include
- ☐ Parameter sharing
 - ☐ Translation invariance
 - ☐ Local connectivity
 - ☐ Sequential processing
47. Which properties allow transformers to handle long-range dependencies?
- ☐ Self-attention
 - ☐ Multi-head attention
 - ☐ Positional encoding
 - ☐ Convolution layers
48. Which factors influence optimization stability?
- ☐ Learning rate
 - ☐ Initialization
 - ☐ Batch normalization
 - ☐ Gradient clipping

49. Which mechanisms help reduce overfitting?

- ☐ Dropout
- ☐ Weight decay
- ☐ Early stopping
- ☐ Increasing model capacity infinitely

50. (4 points) (Numerical) Consider a Large Language Model (LLM) with exactly 1×10^9 (1 Billion) parameters. You are optimizing memory usage during training by utilizing a mixed-precision and quantized setup. Specifically, you store the active model weights in 16-bit precision, compute the gradients in 32-bit precision, and use an 8-bit quantized Adam optimizer (which stores two 8-bit states per parameter).

Calculate the strict total memory required, in Gigabytes (GB), to store these three specific components: the 16-bit weights, the 32-bit gradients, and the 8-bit Adam optimizer states. (Assume $1 \text{ GB} = 10^9$ bytes for simplicity).

Answer: _____

51. (3 points) (MSQ) Gradient clipping is a common technique used during the training of deep neural networks, particularly Recurrent Neural Networks (RNNs). Which of the following statements regarding gradient clipping are true?

- A. It is primarily used to mitigate the vanishing gradient problem by setting a minimum threshold for gradients.
- B. It prevents the exploding gradient problem, which can cause numerical instability and massive updates that destroy learning.
- C. "Clipping by norm" scales the entire gradient vector down if its L_2 norm exceeds a threshold, preserving the direction of the gradient.
- D. "Clipping by value" clamps individual gradient components to a specific range (e.g., $[-c, c]$), which can change the overall direction of the gradient vector.
- E. Gradient clipping directly modifies the model's weights before the optimizer step is calculated.
- F. If the clipping threshold τ is set too low, the model may suffer from artificially induced vanishing gradients, drastically slowing down training.

52. (2 points) (Numerical) In a binary classification task dealing with severe class imbalance (e.g., fraud detection), a weighted binary cross-entropy loss is applied:

$$\mathcal{L} = -[w \cdot y \ln(p) + (1 - y) \ln(1 - p)]$$

where $y \in \{0, 1\}$ is the ground truth label, p is the predicted probability of the positive class, and w is the weight assigned to the positive class.

For a specific training example, the ground truth is positive ($y = 1$), and the model outputs a predicted probability $p = e^{-3}$ (approximately 0.05). If the positive class weight is set to $w = 4$, calculate the exact loss $\mathcal{L}$ for this example.

Answer: _____

53. (3 points) (MSQ) When training models on Natural Language Processing (NLP) tasks, batches often contain sequences (sentences) of varying lengths. How is this disparity handled during training computation?
- A. Shorter sequences are padded with a special <PAD> token to match the length of the longest sequence in the batch or a fixed maximum length.
 - B. The model calculates and backpropagates loss for all tokens, including padding tokens, to penalize the generation of overly short sentences.
 - C. Attention masks (e.g., matrices of 1s and 0s) are utilized to prevent the self-attention mechanism from incorporating information from <PAD> tokens.
 - D. Gradients computed from <PAD> tokens are generally ignored or zeroed out during the loss calculation.
 - E. Modern training loops usually require sorting the entire dataset so that every batch contains sequences of the exact same length, entirely eliminating the need for padding.
 - F. Some tokens are dropped randomly from larger sequences until they all are of same length
54. (2 points) (MCQ) A single-layer perceptron cannot solve XOR because:
- A. Insufficient training data
 - B. Non-convex loss
 - C. Linear decision boundary limitation
 - D. Vanishing gradients
55. (2 points) (MCQ) If all weights initialized to same constant, what happens?
- A. Faster convergence
 - B. Overfitting
 - C. Symmetry problem
 - D. Gradient explosion

56. (2 points) (MCQ) Adam reduces to RMSProp when:
- A. $\beta_1 = 0$
 - B. $\beta_2 = 0$
 - C. learning rate = 0
 - D. $\beta_1 = 1$
57. (2 points) (MCQ) Which causes exploding gradients in RNN?
- A. Small weights
 - B. Eigenvalues > 1
 - C. ReLU
 - D. BatchNorm
58. (3 points) (MSQ) Which help reduce vanishing gradients?
- A. ReLU
 - B. Skip connections
 - C. Sigmoid
 - D. He initialization
59. (3 points) (MSQ) Which are true about optimization methods?
- A. Momentum can overshoot minima
 - B. Adam always converges faster than SGD
 - C. RMSProp avoids infinite decay of learning rate
 - D. Newton method depends on Hessian
60. (3 points) (MSQ) Which operations increase receptive field?
- A. Dilated convolution
 - B. Strided convolution
 - C. Increasing channels
 - D. Pooling
61. (3 points) (MSQ) Which are correct about regularization?
- A. Dropout approximates ensemble
 - B. Early stopping prevents overfitting
 - C. Label smoothing increases entropy
 - D. Data augmentation increases parameters
62. (3 points) Gradient clipping: norm=20, threshold=5 $\rightarrow$ scaling factor?